
MODEL SELECTION AND VALIDATION REPORT (remove_9_biomass model)

OBSERVATIONS:1681

SANITY: OK - all models in the selection process passed sanity checks.

SELECTED RANDOM TERM: spatiotemporal_random_fields_ar1

MODEL SUMMARY (remove_9_biomass model)

Spatiotemporal model fit by REML ['sdmTMB']

Formula: fixed_formula

Time column: year

Family: lognormal(link = 'log')

Conditional model:

	coef.est	coef.se
(Intercept)	5.15	0.18
remove_9_kde_fric	-0.36	0.03
remove_9_kde_feve	-0.36	0.03
sbt_mean	0.34	0.09
sbt_sd	0.02	0.05
sst_sd	0.03	0.04
log_chla_mean	0.08	0.04
depth	0.06	0.05
log_sum_fishing	0.03	0.03

Dispersion parameter: 1.01

Spatiotemporal AR1 correlation (rho): 0.90

Matérn range: 214.06

Spatiotemporal marginal AR1 SD: 0.85

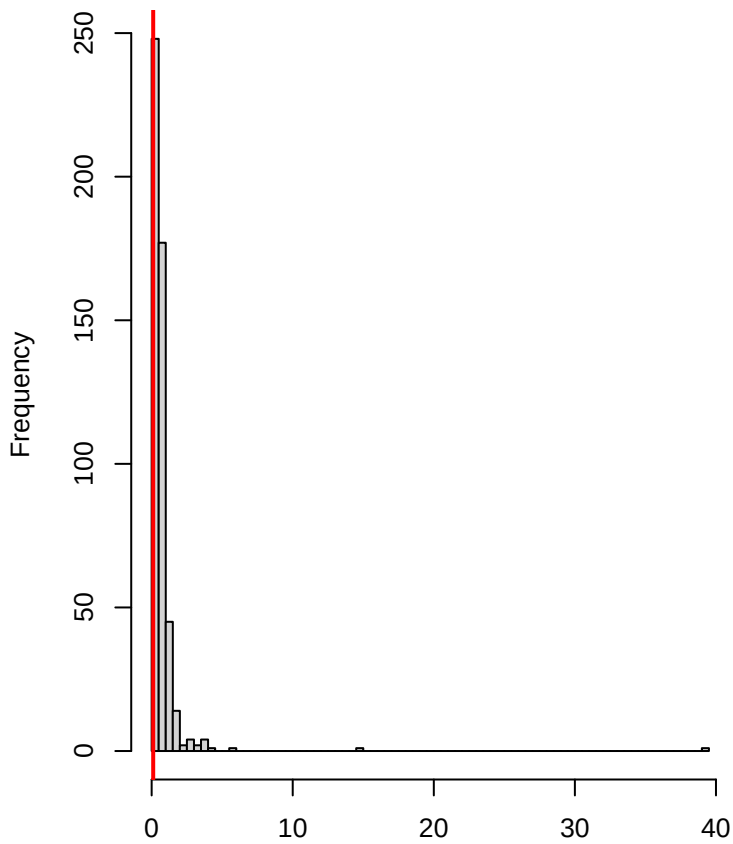
REML criterion at convergence: 10339.283

See ?tidy.sdmTMB to extract these values as a data frame.

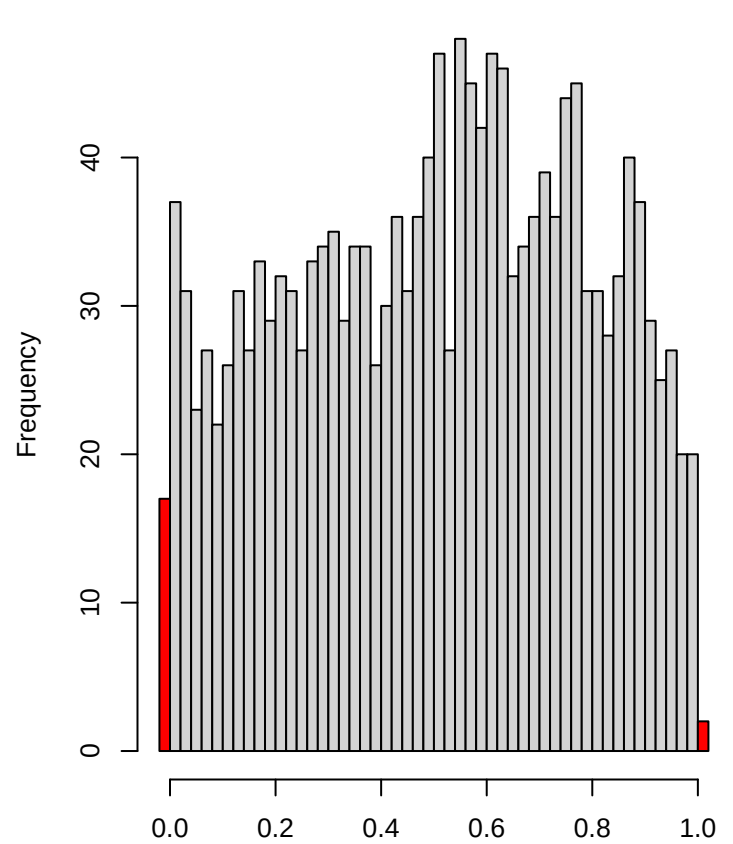
SELECTION STATISTICS

	df	AIC	delta_AIC
spatiotemporal_random_fields_ar1	13	20704.57	0.9276505
spatiotemporal_random_fields_ar1_with_anisotropy	15	20705.49	99.3512862
spatial_random_fields	12	20804.85	3.7411886
spatial_random_fields_with_anisotropy	14	20808.59	46.0599554
spatiotemporal_random_fields_iid_with_anisotropy	14	20854.65	1.6249737
spatiotemporal_random_fields_iid	12	20856.27	136.5327370
error_only	10	20992.80	NA

DHARMa nonparametric dispersion test via sd of residuals fitted vs. simulated



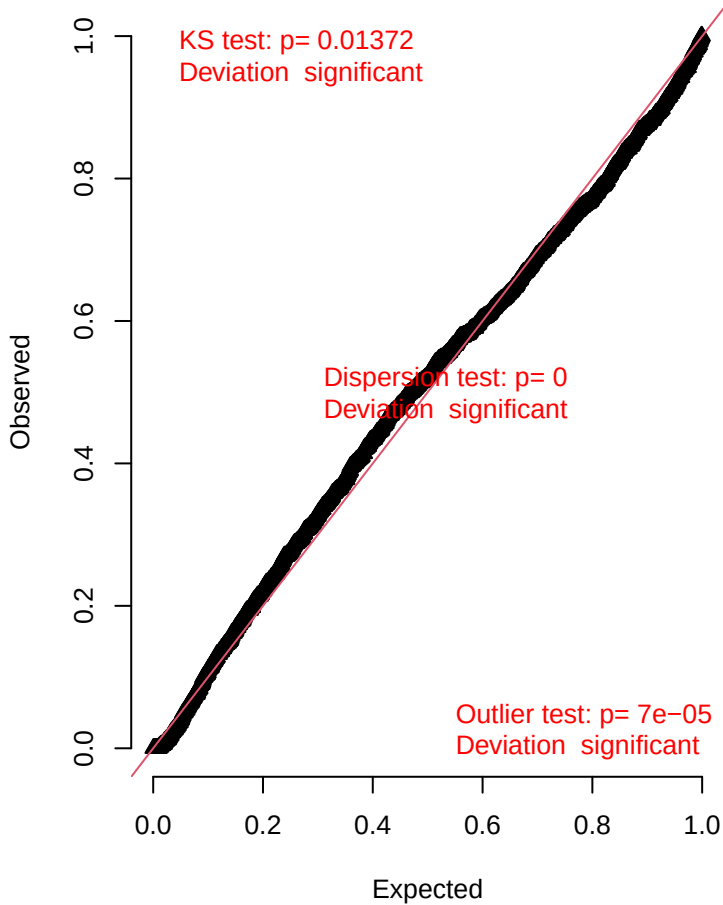
Outlier test significant



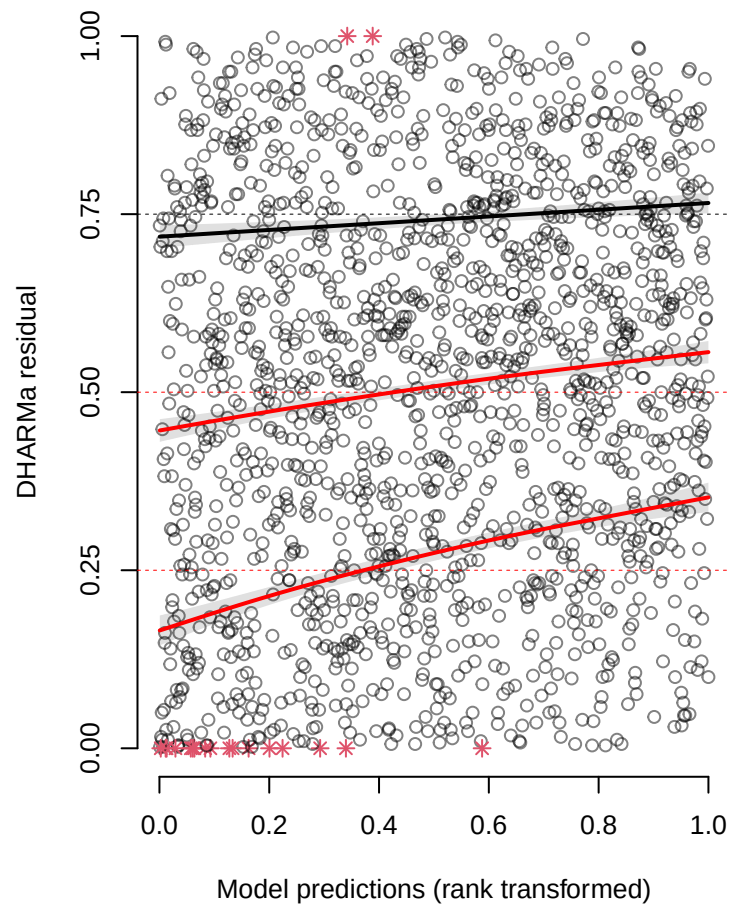
Simulated values, red line = fitted model. p-value (two.sided)

Residuals (outliers are marked red)

QQ plot residuals



DHARMa residual vs. predicted
Quantile deviations detected (red curves)
Combined adjusted quantile test significant



DHARMa Moran's I test for distance-based autocorrelation

Moran's I = 0.022 | H1: Distance-based autocorrelation | p-value = 0.015

